Uncertainty Quantification, Bayesian Inversion & Physics-based Machine learning

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Learning goals

By the end of this lecture, the aim is to understand:

- why uncertainty quantification is central in geophysical inversion,
- the probability essentials needed for Bayesian inversion,
- how Bayes' rule turns the Lecture 6 misfit into a posterior distribution,
- why direct grid or Monte Carlo integration becomes expensive,
- what a Markov chain is and why stationarity matters,
- how the Metropolis–Hastings algorithm samples the posterior,
- how to read traces, marginals, predictive uncertainty, and basic MCMC diagnostics,
- why neural networks are becoming relevant in inversion research.

Intuitive goal

Move from one best model to a probability distribution over plausible models.

Recap and bridge from Lecture 6

In Lecture 6 we used:

- Newton and Gauss–Newton: local, derivative-based optimisation,
- PSO and VFSA: derivative-free global search,
- multiple stochastic runs: a practical way to see families of acceptable models.

What was missing

An optimisation ensemble can show spread, but it is not automatically a sample from a well-defined probability distribution.

- Bayesian inversion defines the target distribution: $p(\mathbf{m} \mid \mathbf{d}^{\text{obs}})$.
- MCMC gives us a practical way to sample that distribution when direct calculation is too expensive.

Why uncertainty matters in geophysics

A geophysical answer is rarely just one model. Decisions depend on the spread of plausible models.

Examples

- How deep is the reservoir?
- How thick is the conductive layer?
- Is an anomaly shallow and small, or deep and large?
- Which features are required by the data?

Sources of uncertainty

- noisy data,
- sparse or uneven coverage,
- non-unique forward responses,
- imperfect physics and parameterization.

Bayesian question

Which Earth models are plausible, and how plausible is each one, given the data and assumptions?

What Is a Posterior?

In Bayesian inversion, a posterior is our updated state of knowledge after seeing the data.

Plain-language idea

- Before seeing the data, we have prior beliefs about plausible models.
- The data then favor some models and rule out others.
- The posterior tells us which models remain plausible after combining both pieces of information.

Informal reading

Posterior = what we believe now, after using the observations.

Three levels of answer

What is the model?

- Level 1 – one model: $\mathbf{m}^{\text{est}}$ from a deterministic inversion.
- Level 2 – an ensemble: several acceptable models from repeated stochastic searches.
- Level 3 – a posterior: a probability distribution $p(\mathbf{m} \mid \mathbf{d}^{\text{obs}})$ over models.

Interpretation

The posterior is stronger than a collection of good models because it attaches probability to model space.

- Marginals tell us uncertainty in individual parameters.
- Joint distributions reveal trade-offs and correlations.
- Posterior predictions show the range of data responses still consistent with the observations.

Start with a simple measurement: height

Suppose we travel across Finland and record the heights of many people:

$$171\text{ cm},\ 182\text{ cm},\ 176\text{ cm},\ 168\text{ cm},\ 174\text{ cm},\ \ldots$$

- With only a few measurements, the values feel random.
- After many measurements, a pattern appears.
- Most people cluster near a middle value.
- Very short and very tall heights are less common.

Key intuition

Uncertainty does not mean no structure. It means we do not know the exact value, but values occur with different frequencies.

From repeated measurements to a PDF

If we sort many heights into bins, we get counts.

Example counts

165–170 cm	1800
170–175 cm	2600
175–180 cm	2300
180–185 cm	1300

Interpretation

- Divide by the total number of people.
- Counts become fractions.
- Fractions approximate probabilities.

Continuous-variable shift

For height, the probability of one exact value is zero. Probability lives on intervals:

$$P(170 \leq H \leq 175).$$

What a probability density function means

A random variable assigns numerical values to uncertain outcomes. For a continuous variable, we use a probability density function (PDF).

Continuous PDF

$$p_H(h) \geq 0, \qquad \int_{-\infty}^{\infty} p_H(h)\, dh = 1.$$

Most important idea

The height of the curve is not probability. Probability is area under the curve:

$$P(a \leq H \leq b) = \int_a^b p_H(h)\, dh.$$

- High density means nearby values are common.
- Low density means nearby values are rare.
- Total area must be one because all possibilities must be somewhere.

Why a bell curve appears so often

Human height is affected by many small contributions:

- genetics,
- nutrition,
- childhood health,
- random biological variation.

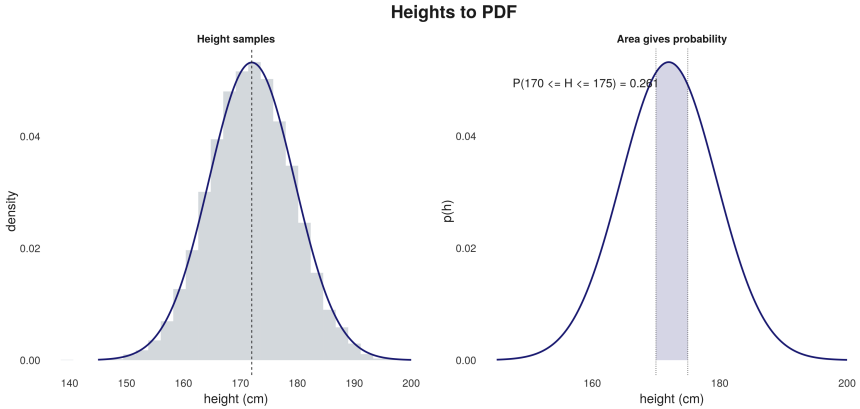
When many small effects combine, a bell-shaped distribution often appears. This is the Gaussian or normal distribution:

$$p(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

Meaning of the parameters

- μ: center or mean.
- σ: spread or standard deviation.

What Height Data Look Like



Demo file

`L7/01-height-pdf-intuition.jl` shows how a histogram, a smooth PDF, and interval probability connect.

Now add a second measurement: weight

Suppose each person gives us a pair of numbers:

$$(\text{height}, \text{weight}).$$

- We no longer ask how common one value is.
- We ask how common combinations are.
- Some combinations are plausible: tall and heavy, short and light.
- Some combinations are less common: very tall and very light.

New object

For two variables, probability is described by a joint PDF $p(h, w)$.

Joint PDFs, marginals, and conditionals

The joint PDF tells us where combinations of height and weight are concentrated.

Marginalization

If we stop caring about weight and only want height, we sum over all possible weights:

$$p(h) = \int p(h, w)\, dw.$$

Likewise,

$$p(w) = \int p(h, w)\, dh.$$

Conditional probability

If height is already known, the probability of weight becomes

$$p(w \mid h) = \frac{p(h, w)}{p(h)}, \qquad p(h) > 0.$$

Dependence, correlation, and trade-offs

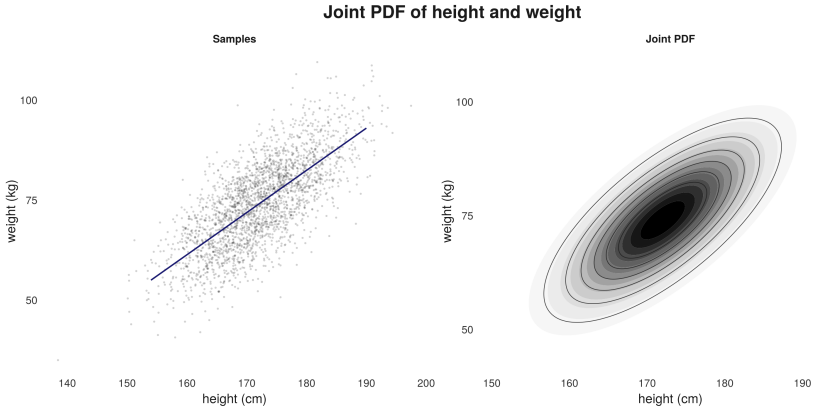
Height and weight are usually related, so the joint PDF is often tilted rather than circular.

- If two variables are independent, then $p(h, w) = p(h)p(w)$.
- If they are dependent, knowing one changes what values of the other remain plausible.
- In inverse problems, this becomes a parameter trade-off.

Why this matters later

Posterior trade-offs in geophysics are the same idea: one parameter can increase while another decreases, and the data still fit.

How Height and Weight Vary Together



Demo file

`L7/02-height-weight-joint-pdf.jl` shows samples, the joint density, and how correlation tilts the high-probability region.

From People to Earth Models

In geophysics, the uncertain quantities are not height and weight. They are model parameters such as

- velocity,
- density,
- conductivity,
- layer thickness.

Same probability ideas, different variables

- one parameter $\rightarrow$ one marginal PDF,
- two parameters $\rightarrow$ one joint PDF,
- many parameters $\rightarrow$ a high-dimensional probability distribution.

Bridge to Bayes

The goal of Bayesian inversion is to describe which Earth models are plausible after we have observed the data.

Bayes' rule

General form

$$p(\mathbf{x} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \mathbf{x})p(\mathbf{x})}{p(\mathbf{y})}.$$

Inverse-problem form

$$p(\mathbf{m} \mid \mathbf{d}^{\text{obs}}) = \frac{p(\mathbf{d}^{\text{obs}} \mid \mathbf{m})p(\mathbf{m})}{p(\mathbf{d}^{\text{obs}})}.$$

- $p(\mathbf{m} \mid \mathbf{d}^{\text{obs}})$: posterior distribution over models.
- $p(\mathbf{d}^{\text{obs}} \mid \mathbf{m})$: likelihood, how well model $\mathbf{m}$ predicts the data.
- $p(\mathbf{m})$: prior information before observing the data.
- $p(\mathbf{d}^{\text{obs}})$: evidence or normalizing constant.

The evidence

The evidence is obtained by integrating over all possible models:

$$p(\mathbf{d}^{\text{obs}}) = \int p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}) p(\mathbf{m})\, d\mathbf{m}.$$

Why this term matters

It normalizes the posterior so that

$$\int p(\mathbf{m} \mid \mathbf{d}^{\text{obs}})\, d\mathbf{m} = 1.$$

- For parameter estimation, we often use the unnormalized posterior

$$p(\mathbf{m} \mid \mathbf{d}^{\text{obs}}) \propto p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}) p(\mathbf{m}).$$

- For model comparison, the evidence itself can matter.
- In high dimensions, the evidence integral is usually the hard part.

Gaussian likelihood for geophysical data

Assume

$$\mathbf{d}^{\text{obs}} = \mathbf{g}(\mathbf{m}) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{C_d}).$$

Then

$$p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}) = \frac{1}{(2\pi)^{N/2}|\mathbf{C_d}|^{1/2}} \exp\left[-\frac{1}{2}\left(\mathbf{d}^{\text{obs}} - \mathbf{g}(\mathbf{m})\right)^T \mathbf{C_d}^{-1}\left(\mathbf{d}^{\text{obs}} - \mathbf{g}(\mathbf{m})\right)\right].$$

Lecture 6 connection

$$\chi_d^2(\mathbf{m}) = \left(\mathbf{d}^{\text{obs}} - \mathbf{g}(\mathbf{m})\right)^T \mathbf{C_d}^{-1}\left(\mathbf{d}^{\text{obs}} - \mathbf{g}(\mathbf{m})\right), \qquad \log p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}) = \text{const} - \frac{1}{2}\chi_d^2(\mathbf{m}).$$

Priors in inverse problems

A prior describes model information before the current data are used.

Bounded uniform prior

$$p(\mathbf{m}) = \begin{cases} \text{constant}, & \mathbf{m} \in \mathcal{M}, \\ 0, & \mathbf{m} \notin \mathcal{M}. \end{cases}$$

Useful when the main prior information is a physically plausible range.

Gaussian prior

$$p(\mathbf{m}) \propto \exp\left[-\frac{1}{2}(\mathbf{m} - \langle \mathbf{m} \rangle)^T \mathbf{C}_{\mathbf{m},\text{prior}}^{-1}(\mathbf{m} - \langle \mathbf{m} \rangle)\right].$$

Useful when a reference model and covariance are meaningful.

MAP estimate and the Lecture 6 objective

The maximum a posteriori model is

$$\mathbf{m}_{\mathrm{MAP}} = \arg\max_{\mathbf{m}} p(\mathbf{m} \mid \mathbf{d}^{\mathrm{obs}}) = \arg\min_{\mathbf{m}} \left[-\log p(\mathbf{d}^{\mathrm{obs}} \mid \mathbf{m}) - \log p(\mathbf{m}) \right].$$

With Gaussian data errors and a Gaussian prior,

$$-\log p(\mathbf{m} \mid \mathbf{d}^{\mathrm{obs}}) = \frac{1}{2} \left(\mathbf{d}^{\mathrm{obs}} - \mathbf{g}(\mathbf{m}) \right)^T \mathbf{C}_{\mathbf{d}}^{-1} \left(\mathbf{d}^{\mathrm{obs}} - \mathbf{g}(\mathbf{m}) \right) + \frac{1}{2} (\mathbf{m} - \langle \mathbf{m} \rangle)^T \mathbf{C}_{\mathbf{m},\mathrm{prior}}^{-1} (\mathbf{m} - \langle \mathbf{m} \rangle) + \mathrm{const.}$$

Key bridge

The regularized least-squares objective from Lecture 6 is the negative log-posterior under Gaussian assumptions.

How can we compute the posterior?

Monte Carlo idea

Draw many model samples from the prior:

$$\mathbf{m}_i \sim p(\mathbf{m}), \qquad i = 1, \ldots, N.$$

Each sample $\mathbf{m}_i$ is one possible Earth model before seeing the data.

- Then evaluate how well each sample explains the data through the likelihood $p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}_i)$.
- Samples that fit the data well should matter more.
- Samples that fit badly should matter less.

How do the weights work?

Because the models were drawn from the prior, the data only need to reweight them:

$$w_i \propto p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}_i), \qquad \sum_i w_i = 1.$$

Meaning of w_i

w_i tells us how much posterior probability is assigned to sample $\mathbf{m}_i$.

- Large w_i: this model stays plausible after seeing the data.
- Small w_i: this model is strongly disfavored by the data.
- The weighted sample cloud is a discrete approximation of the posterior.

How do we get model parameters from the weights?

Once we have samples $\mathbf{m}_i$ and weights w_i, we can estimate model quantities.

Posterior mean model

$$\langle \mathbf{m} \rangle_{\mathrm{post}} \approx \sum_{i=1}^{N} w_i \, \mathbf{m}_i .$$

For one parameter, for example m_k,

$$\langle m_k \rangle \approx \sum_{i=1}^{N} w_i \, m_{i,k} .$$

- This gives an average model, weighted by posterior plausibility.
- We can also compute credible intervals and marginal PDFs from the same weighted samples.
- If we want one best sample, we can take the sample with the largest weight.
- For predicted data, use the same idea: $\sum_i w_i \, \mathbf{g}\left(\mathbf{m}_i\right)$.
- So the goal is not only one model, but a weighted family of plausible models.

Why MCMC?

Direct methods waste many evaluations in regions with tiny posterior probability.

MCMC idea

Construct a dependent sequence of models

$$\mathbf{m}^{(1)}, \mathbf{m}^{(2)}, \ldots, \mathbf{m}^{(N)}$$

whose long-run distribution is the target posterior $p(\mathbf{m} \mid \mathbf{d}^{\text{obs}})$.

- The samples are correlated, not independent.
- But if the chain is built correctly and run long enough, the sample cloud represents the posterior.
- We can then estimate marginal PDFs, means, credible intervals, correlations, and predictive uncertainty.

What is a Markov chain?

A Markov chain is a random process where the next step depends only on where we are now, not on the full history.

Finite-state example

For states Sunny, Cloudy, Rainy, the transition matrix can be

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}.$$

- Each row tells us what tomorrow's weather could be, given today's weather.
- For example, if today is Sunny, the first row says tomorrow is most likely Sunny again, but Cloudy or Rainy are also possible.
- The important idea is memory of just one step: today influences tomorrow.

Stationary distribution: weather-chain example

Stationary distribution

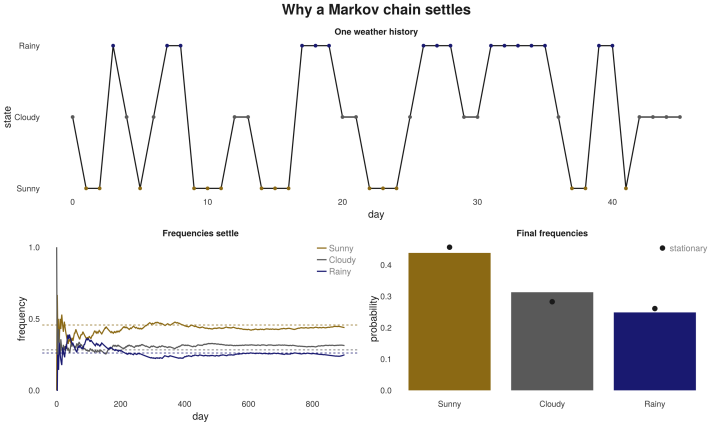
A stationary distribution is the long-run balance of time spent in each state.

Why this matters for MCMC

In MCMC we design the transition rule so that the posterior is the stationary distribution.

- Imagine running the weather model for a very long time.
- At first, the proportions depend on the starting day.
- After many steps, those proportions settle to stable values.
- Those stable values are the stationary distribution.
- In the weather example, the animation shows these running frequencies settling down.

A Simple Weather Chain



Demo file

`L7/04-markov-chain-demo.jl` simulates the Sunny–Cloudy–Rainy chain, saves a simpler figure, and writes a GIF showing frequencies converging toward the stationary distribution.

MCMC target distribution

For Bayesian inversion, the target density is the posterior:

$$\pi(\mathbf{m}) = p(\mathbf{m} \mid \mathbf{d}^{\text{obs}}) \propto p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}) p(\mathbf{m}).$$

Important practical point

Metropolis–Hastings only needs the posterior up to a proportionality constant. The evidence $p(\mathbf{d}^{\text{obs}})$ cancels in the acceptance ratio.

- We need to evaluate $\log p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}) + \log p(\mathbf{m})$.
- We do not need to compute the full evidence integral.
- This is one reason MCMC is so useful for inverse problems.

Metropolis–Hastings algorithm

Both methods follow the same outer loop: current model → proposal → accept or reject.

VFSA reminder

Goal: find a low-misfit model.

$$P_{\mathrm{acc}} = \begin{cases} 1, & \Delta\Phi \leq 0, \\ \exp\left(-\dfrac{\Delta\Phi}{T}\right), & \Delta\Phi > 0. \end{cases}$$

- Draw $u \sim U(0,1)$ and accept if $u \leq \alpha$.
- If rejected, repeat the current model.
- Main difference: VFSA is an optimizer; MH is a sampler.

Metropolis–Hastings

Goal: sample the posterior.
Proposal: $\mathbf{m}' \sim q(\mathbf{m}' \mid \mathbf{m}_i)$.

$$\alpha = \min\left[1, \frac{\pi(\mathbf{m}')q(\mathbf{m}_i \mid \mathbf{m}')}{\pi(\mathbf{m}_i)q(\mathbf{m}' \mid \mathbf{m}_i)}\right].$$

Use log probabilities

In practice we compute with log probabilities to avoid numerical underflow.

General log acceptance

$$\log \alpha = \min \left[0,\ \log \pi(\mathbf{m}') - \log \pi(\mathbf{m}_i) + \log q(\mathbf{m}_i \mid \mathbf{m}') - \log q(\mathbf{m}' \mid \mathbf{m}_i)\right].$$

Acceptance test

Draw $u \sim U(0,1)$ and accept if

$$\log u < \log \alpha.$$

The Posterior Used in This Example

The demo uses the Lecture 6 hard forward model:

$$d_i^{\mathrm{pre}} = m_1 \exp\left[-\left(\frac{t_i}{m_2}\right)^2\right] + m_3.$$

The prior is bounded uniform:

$$\mathbf{m} \in [m_1^{\min}, m_1^{\max}] \times [m_2^{\min}, m_2^{\max}] \times [m_3^{\min}, m_3^{\max}].$$

Log posterior

$$\log \pi(\mathbf{m}) = \begin{cases} -\dfrac{1}{2\sigma_d^2}\left\|\mathbf{d}^{\mathrm{obs}} - \mathbf{g}(\mathbf{m})\right\|_2^2 + \text{const}, & \mathbf{m} \text{ inside bounds}, \\ -\infty, & \mathbf{m} \text{ outside bounds}. \end{cases}$$

Full MCMC inversion algorithm

Given observed data $\mathbf{d}^{\text{obs}}$, a prior $p(\mathbf{m})$, a likelihood $p(\mathbf{d}^{\text{obs}} \mid \mathbf{m})$, and a starting model $\mathbf{m}_0$:

- Choose the number of samples, burn-in length, and a proposal rule $q(\mathbf{m}' \mid \mathbf{m}_i)$.
- Start from the current model $\mathbf{m}_i$ and evaluate its log posterior

$$\log \pi(\mathbf{m}_i) = \log p(\mathbf{d}^{\text{obs}} \mid \mathbf{m}_i) + \log p(\mathbf{m}_i).$$

- Propose a trial model $\mathbf{m}' \sim q(\mathbf{m}' \mid \mathbf{m}_i)$.
- Evaluate the trial log posterior $\log \pi(\mathbf{m}')$.
- Compute the log acceptance probability

$$\log \alpha = \min \left[0,\ \log \pi(\mathbf{m}') - \log \pi(\mathbf{m}_i) + \log q(\mathbf{m}_i \mid \mathbf{m}') - \log q(\mathbf{m}' \mid \mathbf{m}_i)\right].$$

- Draw $u \sim U(0,1)$. If $\log u < \log \alpha$, accept the proposal; otherwise keep the current model.
- Store the resulting sample and repeat until the chain is long enough.
- Discard burn-in, then use the remaining samples to compute marginals, means, intervals, and predictive uncertainty.

Output

The result is not one best model, but a sample-based approximation of the posterior distribution over plausible models.

Sampling the Posterior

The MCMC demo and all lecture code are in the repository

`https://github.com/pankajkmishra/GEOM2057`

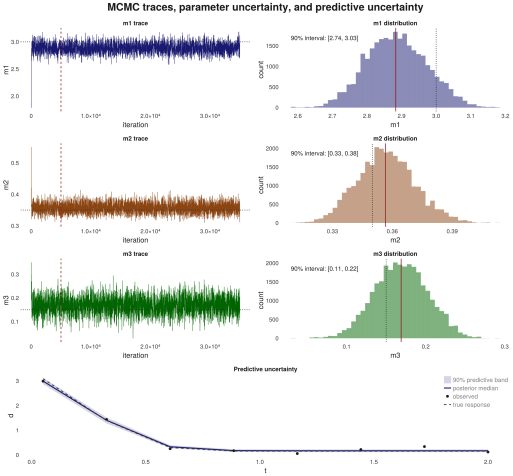
Main files

- `L6/05-problem-setup-hard.jl`: hard forward model and objective helpers.
- `L6/06-data-generation-hard.jl`: repeatable sparse/noisy synthetic data.
- `L7/05-metropolis-hastings-geophysical.jl`: posterior sampling and CairoMakie plots.

Output

The output shows trace behavior, parameter uncertainty, and predictive uncertainty.

One MCMC Result



How to read an MCMC result

Trace plots

- Check whether the chain has moved away from its starting point.
- Look for poor mixing, long drifts, or chains stuck in one region.

Marginal histograms

- Summarize uncertainty in one parameter at a time.
- Report posterior median, mean, and credible intervals.

Predictive uncertainty

- Push posterior samples through the forward model.
- The predictive band shows which data responses remain plausible.

Burn-in, correlation, and effective samples

MCMC samples are not independent draws.

- Burn-in: early samples may reflect the starting model more than the posterior.
- Autocorrelation: nearby samples in the chain are usually similar.
- Effective sample size: the number of independent samples that would contain similar information.
- Multiple chains: starting from different regions helps reveal non-convergence.

Practical warning

A long chain is not automatically a good chain. We still need diagnostics and geological judgment.

What MCMC gives and what it does not

MCMC gives

- posterior samples,
- parameter uncertainty,
- parameter trade-offs,
- posterior predictive checks.

MCMC does not automatically give

- a perfect answer from a short run,
- independence between samples,
- protection from a wrong likelihood,
- protection from a bad prior or bad forward model.

Main message

Bayesian inversion quantifies uncertainty conditional on the assumptions. If the assumptions change, the posterior can change too.

Common MCMC extensions

Metropolis–Hastings is the simplest useful entry point. Many extensions keep the same goal but improve efficiency or robustness.

Examples

- **Gibbs sampling**: sample from conditional distributions when they are available.
- **Adaptive Metropolis**: adjust proposal covariance during warm-up.
- **Parallel tempering**: run chains at different temperatures to explore multi-modal targets.
- **Sequential Monte Carlo**: move a population of weighted particles through a sequence of distributions.
- **Ensemble samplers**: use an ensemble of walkers to adapt proposal directions.

Core idea

Build a chain whose stationary distribution is the posterior.

Why inversion looks like training

Geophysical inversion

- Start with unknown model parameters.
- Predict data with a forward model.
- Compare predicted and observed data.
- Update parameters to reduce the mismatch.

Neural-network training

- Start with unknown weights.
- Predict outputs with the network.
- Compare predicted and target outputs.
- Update weights to reduce the loss.

Big similarity

Both problems search for parameters by repeatedly predicting, comparing, and updating.

Can neural networks help inversion?

Yes. There are many ways to use neural networks in inversion, and this is now a very active research area.

One direction we are exploring

Physics-informed machine learning via implicit neural representation (INR).
Instead of storing one value in each cell, we train a neural network that maps coordinates to Earth properties.

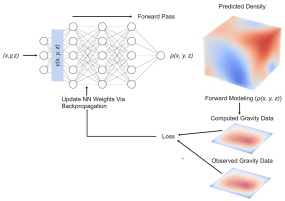


Figure 1: Physics-based machine learning inversion framework

Read more

Please read *Three-dimensional inversion of gravity data using implicit neural representations and scientific machine learning* (arXiv:2510.17876). The paper includes Python codes.

So what did we learn?

- Uncertainty matters because many Earth models can explain noisy, incomplete data.
- Bayes' rule combines likelihood and prior information into a posterior distribution.
- With Gaussian data errors, the Lecture 6 least-squares misfit is the negative log-likelihood up to a constant.
- With a Gaussian prior, regularization becomes a prior penalty in the negative log-posterior.
- Grid search and plain Monte Carlo are transparent but scale poorly with dimension.
- A Markov chain has a long-run stationary distribution.
- Metropolis–Hastings constructs a chain whose stationary distribution is the posterior.
- MCMC output should be read through traces, marginals, and diagnostics.
- Geophysical inversion and neural-network training share the same predict-compare-update structure.
- Neural networks can be used for inversion in many ways, and this is a popular research direction.
- One example is physics-informed machine learning with implicit neural representations.

Take-home message

Bayesian inversion helps us describe plausible Earth models, and modern machine-learning ideas may give us new ways to represent and solve these problems.